

# HAMILTONIAN ELEMENTS IN ALGEBRAIC K-THEORY

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ABSTRACT. A Hamiltonian bundle  $M \hookrightarrow P \rightarrow X$  (with monotone compact fibers) induces via Floer theory a type of “bundle of  $A_\infty$  categories” over  $X$ , with fiber given by the Fukaya category of  $M$ . Morita theory of  $A_\infty$  categories, the above picture for  $X = S^m$ , and geometric representation theory yield the following: if  $G$  is a compact Lie group and  $R$  is a commutative ring then there is a natural group homomorphism  $\pi_m(BG) \rightarrow K_m^{Cat}(R)$ , where  $K_m^{Cat}(R)$  are a type of categorified algebraic  $K$ -theory groups of  $R$ , analogous to Toën’s secondary  $K$ -theory. Furthermore, the homomorphisms are induced by a natural homotopy class  $[B\text{Ham}(G/T) \rightarrow K^{Cat}(R)]$ , where  $K^{Cat}(R)$  is the associated spectrum. We also construct underlying maps of this type to classical algebraic  $K$ -theory of  $R$ . This framework gives a geometry-powered proof that  $K_2^{Cat}(\mathbb{Z})$  is infinitely generated (with the details to appear in a future work). This is in contrast to Quillen’s finite generation result for standard algebraic  $K$ -theory of  $\mathbb{Z}$ . Taking the Langlands dual of  $G$ , we explore a conjectural relationship between the images of the corresponding homomorphisms above.

## 1. INTRODUCTION

A guiding idea is that a Hamiltonian bundle  $M \hookrightarrow P \rightarrow Y$  gives rise to a kind of bundle of  $A_\infty$  categories, with fiber the Fukaya category of  $M$ . Precisely what this means, is an interesting mathematical problem in its own right, and one answer is developed in [20]. Notably, these are not local systems; as shown in [19], the resulting bundles are generally non-trivial over spheres.

The second idea, due to Toën [24], is that there is a refinement of algebraic  $K$ -theory of a commutative ring  $R$ , by replacing  $R$ -modules with  $dg$  or  $A_\infty$  categories over  $R$ . We show that these two ideas naturally fit together. We use the above bundle assignment, with  $Y = S^m$ , to associate to each such  $P$  an element in categorified and standard algebraic  $K$ -theory of  $R$ . In particular, we get a geometric insight into algebraic  $K$ -theory of  $\mathbb{Z}$ , especially its categorified variant  $K_m^{Cat}(\mathbb{Z})$ . For example, we obtain an infinite generation result for  $K_2^{Cat}(\mathbb{Z})$ . At the end we explore connections with Langlands duality. We thus link algebraic  $K$ -theory, Floer theory, and aspects of geometric representation theory in one framework.

Although in the main examples here the principal actors are compact groups, and in particular  $PU(n)$ , the framework is most naturally understood from the perspective of the group of Hamiltonian symplectomorphisms of some symplectic manifolds. We quickly introduce the latter. Let  $(M, \omega)$  be a symplectic manifold, throughout assumed to be compact, with  $\omega$  a closed nondegenerate 2-form on  $M$ . The group of diffeomorphisms  $\phi$  of  $M$ , satisfying  $\phi^*\omega = \omega$ , is called the symplectomorphism group of  $(M, \omega)$ . When  $M$  is simply connected, this is also the group of Hamiltonian symplectomorphisms, which we denote by  $\mathcal{H} = \text{Ham}(M, \omega)$ . In general, a symplectomorphism is Hamiltonian if it is the time-one map of an isotopy generated by a family of vector fields  $\{X_t\}$ ,  $t \in [0, 1]$ , satisfying

$$\omega(X_t, \cdot) = dH_t(\cdot),$$

for some family of smooth functions  $\{H_t\}$  on  $M$ .

The group  $\mathcal{H}$  is one of the more enigmatic objects of mathematics. It is always (unless  $M = pt$ ) an infinite-dimensional Fréchet Lie group with its  $C^\infty$  topology. But it has a natural, bi-invariant

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Finsler metric called the Hofer metric. So in some ways it behaves like a compact group. Its topological and algebraic structure is tied to the main problems of Hamiltonian dynamics, see for instance Polterovich [15].

Much of the algebraic/topological structure of  $\mathcal{H}$  is reflected in the properties of Hamiltonian fibrations, which are the main geometric ingredients in what follows. A *Hamiltonian fibration*  $M \hookrightarrow P \rightarrow Y$  is a smooth fiber bundle over  $Y$ , with fiber a symplectic manifold  $(M, \omega)$ , whose structure group is contained in  $\text{Ham}(M, \omega)$ . For example, take  $(M, \omega) = (\mathbb{CP}^n, \omega_{FS})$ , where  $\omega_{FS}$  is the Fubini-Study symplectic form. We will from now on omit  $\omega_{FS}$  from all notation; it is implicitly always the chosen symplectic form on  $\mathbb{CP}^n$ . And suppose for instance that the structure group of  $\mathbb{CP}^n \hookrightarrow P \rightarrow Y$  is  $G = PU(n+1) \subset \text{Ham}(\mathbb{CP}^n)$ .

The other ingredient is algebraic  $K$ -theory, which is a construction associating an infinite loop space <sup>1</sup>  $K(R)$  to any ring  $R$ ; and so abelian groups:

$$K_m(R) = \pi_m(K(R)), \quad m \in \mathbb{N},$$

called  $K$ -theory groups of  $R$ . This is a powerful tool in algebraic topology and ring theory, with applications for instance in number theory and geometric topology.

One motivating example for algebraic  $K$ -theory is the following. Take  $R$  to be the non-commutative ring  $\mathcal{K}$  of compact operators on a separable complex Hilbert space, the algebraic  $K$ -theory space  $K(\mathcal{K})$  <sup>2</sup> is identified with the topological  $K$ -theory space  $KU = BU \times \mathbb{Z}$ , see Karoubi [9], Suslin-Wodzicki [23]. Thus a complex vector bundle  $E$  over a space  $X$  determines a homotopy class  $[f_E : X \rightarrow K(\mathcal{K})]$ , which we might call a *geometric generator* of the corresponding abelian group. We shall show that in a certain sense this extends to a general ring  $R$ , provided it is commutative, and replacing complex vector bundles by Hamiltonian fibrations.

For each commutative ring  $R$ , we construct an infinite loop space  $K^{Cat}(R)$  called categorified algebraic  $K$ -theory of  $R$ . The groups  $K_m^{Cat}(R) = \pi_m(K^{Cat}(R))$  can also be understood to be generated by the aforementioned bundles of  $A_\infty$  categories over  $S^m$ , with fibers  $A_\infty$  categories with finite type, see Definition 3.1. In principle, this categorified algebraic  $K$ -theory of  $R$  is only a slight variation of the secondary  $K$ -theory of  $R$  introduced by Toën, with the main difference being that  $\mathbb{Z}$ -graded chain complexes are replaced by  $\mathbb{Z}_2$ -graded chain complexes. The following is a conjunction of Theorem 4.5 and Corollary 4.8.

**Theorem 1.1.** *If the Fukaya category of  $M$  is finite type, then for each  $R$  the data of the isomorphism class of  $M \hookrightarrow P \rightarrow Y$  determines a class  $[Y \rightarrow K^{Cat}(R)]$ . Restricting to  $Y = S^m$  and  $M = \mathbb{CP}^n$ , this determines homomorphisms*

$$\phi_m : \pi_m(B \text{Ham}(\mathbb{CP}^n)) \rightarrow K_m^{Cat}(R),$$

and so homomorphisms

$$\phi_m : \pi_m(BPU(n)) \rightarrow K_m^{Cat}(R).$$

*Example 1.* In the case of  $n = 1$ , in [19], using geometric arguments we outline the proof of non-triviality of  $\phi_4$  for  $R = \mathbb{Z}$ . This contrasts with a theorem of Rognes [17] on the triviality of  $K_4(\mathbb{Z})$ .

**Claim 1.2.**

$$\phi_2 : \mathbb{Z}_p \simeq \pi_2(BPU(p)) \rightarrow K_2^{Cat}(\mathbb{Z})$$

is injective for each prime  $p$ . Consequently,  $K_2^{Cat}(\mathbb{Z})$  is infinitely generated.

<sup>1</sup>In this paper, the term infinite loop space is meant to be synonymous with connective  $\Omega$ -spectrum, i.e. a sequence of spaces  $\{E_n\}_{n \in \mathbb{N}}$  with  $E_n \simeq \Omega E_{n+1}$ . Then  $K(R)$  is the notation for the 0-space  $E_0$  of the corresponding spectrum.

<sup>2</sup>Rather one must take a certain non-connective variant of the algebraic  $K$  theory construction.

Since the groups  $K_m(\mathbb{Z})$  are always finitely generated by the foundational results of Quillen [16], the claim is perhaps surprising.

One geometric ingredient for the proof is the now classical part of symplectic geometry: the Seidel homomorphism [21]. The details will appear in a future work.

**1.1. Generalizing to  $G/T$ .** Theorem 1.1 extends to  $M = G/T$ , where  $G$  is compact. However, the details in that case also involve non-elementary geometric representation theory, see Section 4.1. We still get natural maps

$$\pi_m(B\mathrm{Ham}(G/T)) \rightarrow K_m^{\mathrm{Cat}}(R),$$

but after fixing a certain choice (a distinguished toric degeneration of  $G/T$ ). Without that we get into questions of finiteness of the Fukaya category  $\mathrm{Fuk}(G/T)$ .

**1.2. Pushing forward to classical  $K$ -theory.** To partially bypass the need for choices, we can pass to classical algebraic  $K$ -theory, instead of its categorified analogue. We take a 2-periodic variant of classical algebraic  $K$ -theory of  $R$ , which we denote by  $K_m^{\mathbb{Z}_2}(R)$ . Technically,  $K^{\mathbb{Z}_2}(R)$  is the algebraic  $K$ -theory infinite loop space  $K(W)$ , where  $W$  is the Waldhausen category of 2-periodic complexes of  $R$ -modules.

*Remark 1.3.* It was pointed out to me by Bertrand Toën that for  $R$  a regular ring, for example  $R = \mathbb{Z}$ ,  $K_m^{\mathbb{Z}_2}(R) \simeq K_m(R) \oplus K_{m-1}(R)$ . This can be proved with  $A^1$  homotopy theory. Although we don't technically need this.

The same basic framework of the paper yields the following.

**Theorem 1.4.** *For any compact Lie group  $G$ , there is a homotopy natural map:*

$$(1.5) \quad h : \mathrm{BG} \rightarrow K^{\mathbb{Z}_2}(R),$$

*and so there are natural homomorphisms*

$$h_m : \pi_m(\mathrm{BG}) \rightarrow K_m^{\mathbb{Z}_2}(R).$$

Taking the Langlands dual of  $G$ , we explore correspondence of the respective mappings in Section 6.

*Question 1.* Is it possible to obtain the classes

$$[\mathrm{BG} \rightarrow K^{\mathbb{Z}_2}(R)]$$

by techniques of homotopy theory and algebraic geometry? In other words, can we bypass the geometric analysis of the construction?

It should be noted that compared to the maps  $\phi_m$ , showing non triviality of the maps  $h_m$  is harder (and is an open problem), even restricting to  $PU(n)$  and  $R = \mathbb{Z}$ . This might seem paradoxical, but the idea is that  $K_m^{\mathrm{Cat}}(R)$  sees more information, so it is easier to find nontrivial things in the image of  $\phi_m$ .

## 2. PRELIMINARIES

We will mostly follow Keller [10], but the base commutative ring will be denoted  $R$  to avoid overloading notation. An  $A_\infty$  category  $C$  over  $R$  is a form of non-associative dg-category, with  $\mathrm{hom}_C(a, b)$  a cochain complex over  $R$ . We refer the reader to standard references for further details, see for instance [14] which will come in use later.

Denote by  $A_\infty \mathrm{Cat}_R^{\mathrm{strict}}$  the category of strictly unital,  $\mathbb{Z}$ -graded  $A_\infty$  categories with morphisms being strict unital  $A_\infty$  functors. We set  $A_\infty \mathrm{Cat}_R^{\mathbb{Z}_n, \mathrm{strict}}$  to be like  $A_\infty \mathrm{Cat}_R^{\mathrm{strict}}$ ; but with chain complexes

$\mathbb{Z}_n$ -graded instead of  $\mathbb{Z}$ -graded. This means that all structure maps, functors, etc., are with respect to the  $\mathbb{Z}_n$ -grading. Various subscripts  $R$  may be omitted.

There is an endofunctor

$$T : A_\infty \text{Cat}_R^{\mathbb{Z}_n, \text{strict}} \rightarrow A_\infty \text{Cat}_R^{\mathbb{Z}_n, \text{strict}},$$

s.t.  $T(B)$  is idempotent complete and pretriangulated for each  $B$ . We may denote by  $\widehat{C}$  the category  $T(C)$ .

Let  $\text{Cat}_R^{\mathbb{Z}_n}$  denote the category of small  $R$ -linear  $\mathbb{Z}_n$ -graded categories. For  $C \in A_\infty \text{Cat}_R^{\mathbb{Z}_n, \text{strict}}$ , set  $hC \in \text{Cat}_R^{\mathbb{Z}_n}$  to be the category with the same objects and with  $\text{hom}_{hC}(a, b) = H^*(\text{hom}_C(a, b))$ .

Given a morphism  $f : A \rightarrow B$  in  $A_\infty \text{Cat}_R^{\mathbb{Z}_n, \text{strict}}$ , we denote by  $f_* : hA \rightarrow hB$  the induced morphism in  $\text{Cat}_R^{\mathbb{Z}_n}$ . We say that  $f$  is a *quasi-equivalence* if  $f_*$  is an equivalence. We will say that  $f$  is a *Morita equivalence* if the induced functor  $T(A) \rightarrow T(B)$  is a quasi-equivalence. We denote by  $Hmo^{\mathbb{Z}_2}$  the homotopy category of  $A_\infty \text{Cat}_R^{\text{strict}, \mathbb{Z}_2}$  with respect to Morita equivalences, as explained by Keller this is a pointed category with the zero object: the category 0 with one object and zero endomorphism complex.

The category  $A_\infty \text{Cat}_R^{\mathbb{Z}_n, \text{strict}}$  is cocomplete, see [14]. Furthermore, it has a model category structure [13]. Applying left Bousfield localization along Morita equivalences we get what is often called the Morita model structure. Although, we do not explicitly need the language of model structures. The main algebraic ingredient is the following.

**Definition 2.1.** A *short exact sequence* in  $A_\infty \text{Cat}_R^{\text{strict}, \mathbb{Z}_2}$  is a sequence  $A \xrightarrow{I} B \xrightarrow{P} C$ , such that in  $Hmo^{\mathbb{Z}_2}$ ,  $I$  is the kernel of  $P$  and  $P$  is the cokernel of  $I$ .

As explained by Keller in [10], an explicit example for such a short exact sequence is given by a Drinfeld dg-quotient sequence  $A \hookrightarrow B \rightarrow B/A$ .

**2.1. Simplicial sets.** We will need basic theory of simplicial sets, particularly the notion of a simplex category, for the definition of bundles of  $A_\infty$  categories. We denote by  $\Delta$  the simplex category:

- The set of objects of  $\Delta$  is  $\mathbb{N}$ .
- $\text{hom}_\Delta(n, m)$  is the set of non-decreasing maps  $[n] \rightarrow [m]$ , where  $[n] = \{0, 1, \dots, n\}$ , with its natural order.

A simplicial set  $X$  is a functor

$$X : \Delta^{op} \rightarrow \text{Set}.$$

$X(n)$  is the set of  $n$ -simplices of  $X$ .  $\Delta^d$  will denote a particular simplicial set: the standard representable  $d$ -simplex, with

$$\Delta^d(n) = \text{hom}_\Delta(n, d).$$

Morphisms of simplicial sets are given by natural transformations of functors and this forms a category denoted by  $s\text{Set}$ .

An important ingredient in our construction will be the following.

**Definition 2.2.** For  $X$  a simplicial set,  $\Delta(X)$  will denote the *simplex category of  $X$* . This is the category such that:

- The set of objects  $\text{obj } \Delta(X)$  is the set of simplicial maps:

$$\Sigma : \Delta^d \rightarrow X, \quad d \geq 0.$$

- Morphisms  $f : \Sigma_1 \rightarrow \Sigma_2$  are commutative diagrams in  $sSet$ :

$$(2.3) \quad \begin{array}{ccc} \Delta^d & \xrightarrow{\quad} & \Delta^n \\ & \searrow \Sigma_1 & \downarrow \Sigma_2 \\ & & X. \end{array}$$

Given a morphism  $f : X \rightarrow Y$  of simplicial sets, we set  $\Delta f : \Delta(X) \rightarrow \Delta(Y)$  to be the functor defined on objects by  $\Delta f(\Sigma) = f \circ \Sigma$ . If  $m : \Sigma_1 \rightarrow \Sigma_2$  is a morphism in  $\Delta(X)$ :

$$\begin{array}{ccc} \Delta^k & \xrightarrow{m} & \Delta^d \\ & \searrow \Sigma_1 & \downarrow \Sigma_2 \\ & & X, \end{array}$$

then obviously the diagram below also commutes:

$$\begin{array}{ccc} \Delta^k & \xrightarrow{m} & \Delta^d \\ & \searrow h_1 & \downarrow h_2 \\ & & Y, \end{array}$$

where  $h_i = \Delta f(\Sigma_i) = f \circ \Sigma_i$ ,  $i = 1, 2$ . And so the latter diagram determines a morphism  $\Delta f(m) : h_1 \rightarrow h_2$  in  $\Delta(Y)$ . Clearly, this determines a functor  $\Delta f$  as needed.

**Definition 2.4.** Let  $X$  be a simplicial set, and  $C$  any category. Let  $F_i : \Delta(X) \rightarrow C$ ,  $i = 0, 1$  be functors. A **concordance** of  $F_0$  to  $F_1$  is a functor

$$\tilde{F} : \Delta(X \times \Delta^1) \rightarrow C$$

such that  $\tilde{F}|_{\Delta(X \times \{0\})} = F_0$  and  $\tilde{F}|_{\Delta(X \times \{1\})} = F_1$ , where  $\{0\}, \{1\}$  here denote the images of the natural end point inclusions  $\Delta^0 \rightarrow \Delta^1$ .

### 3. DEFINITION OF THE CATEGORIFIED ALGEBRAIC K-THEORY GROUPS OF $R$

We need to impose some algebraic finiteness conditions on our  $A_\infty$  categories. otherwise, any  $K$ -theory we construct would be trivial due to the Eilenberg swindle; the same reason that  $K$ -theory of infinite-dimensional vector bundles is trivial. A natural requirement is that our  $A_\infty$  categories be smooth and proper, or saturated in the sense of Kontsevich-Soibelman [12], as this is a dualizability condition, see [25]. In the  $\mathbb{Z}_n$ -graded setting, the analogue of this condition is more complex, we will instead work with a less restrictive notion, which is not too hard to check for geometric examples.

Set  $\mathcal{A} = \mathcal{A}_R \subset A_\infty \text{Cat}_R^{\mathbb{Z}_2, \text{strict}}$  to be the full subcategory whose objects are finite type, where the latter is defined as follows.

**Definition 3.1.** We say that  $B \in A_\infty \text{Cat}_R^{\mathbb{Z}_2, \text{strict}}$  is **finite type** if the following conditions are satisfied.

- $B$  is Morita equivalent to an  $A_\infty$  algebra, whose underlying chain complex is projective and finitely generated over  $R$  in each degree.
- In the category of 2-periodic complexes over  $R$ , the Hochschild homology complex of  $B$  over  $R$  is quasi-isomorphic to a complex that is projective and finitely generated over  $R$  in each degree.

Set  $\hat{\mathcal{A}}_R \subset \mathcal{A}_R$  to be the full subcategory with objects pretriangulated and idempotent complete. As outlined by Toën in [24],  $\hat{\mathcal{A}}_R \subset \mathcal{A}_R$  has the structure of a Waldhausen category. The cofibrations are taken to be fully-faithful, strict, unital, injective on sets of objects  $A_\infty$  functors. Weak equivalences are Morita equivalences. To be strict, this is not a Waldhausen category but  $\infty$ -Waldhausen in the sense

of Barwick [2], as for example the zero category  $0$  (the category with one object and zero complex as endomorphism set) is not initial in  $\widehat{\mathcal{A}}_R$  but the hom-sets  $\text{hom}_{\widehat{\mathcal{A}}_R}(0, A)$  are contractible when taken in the  $\infty$ -localization of  $\widehat{\mathcal{A}}_R$  with respect to the Morita equivalences.

Given this we define

$$K_m^{Cat}(R) = \pi_m K(\widehat{\mathcal{A}}_R),$$

where  $K(\widehat{\mathcal{A}}_R)$  is the associated Waldhausen spectrum. Calculations from this general perspective are difficult, we now describe a more concrete presentation.

### 3.1. Concrete presentation of $K_m^{Cat}(R)$ .

**Definition 3.2.** Let  $X$  be a simplicial set. A **bundle of  $A_\infty$  categories over  $X$**  is a functor  $F : \Delta(X) \rightarrow w\widehat{\mathcal{A}}_R$ .  $F$  will be called a **bundle functor over  $X$** .

Suppose we have bundle functors over  $X$ , fitting into a sequence of natural transformations of functors to  $\widehat{\mathcal{A}}_R$  (as opposed to  $w\widehat{\mathcal{A}}_R$ ):

$$(3.3) \quad F_0 \rightarrow F_1 \rightarrow F_2,$$

such that for each object  $\Sigma \in \Delta(S^k)$  the resulting sequence

$$(3.4) \quad F_0(\Sigma) \rightarrow F_1(\Sigma) \rightarrow F_2(\Sigma),$$

is a short exact sequence in  $\mathcal{A}_R$ . Then we call  $F_0 \rightarrow F_1 \rightarrow F_2$  a **short exact sequence**.

**Definition 3.5.** Let  $X$  be a Kan complex. Set  $B(X, R)$  to be the free abelian group generated by the set of concordance classes  $[F]$  of functors  $F : \Delta(X) \rightarrow w\mathcal{A}_R$ . We define  $K^{Cat}(X, R)$  to be the quotient of  $B(X, R)$  by the subgroup generated by

$$\{[F_0] - [F_1] + [F_2] \mid \text{there is a short exact sequence } F_0 \rightarrow F_1 \rightarrow F_2\}.$$

This is clearly contravariantly functorial in  $X$  so if  $f : X \rightarrow Y$  is a simplicial map then we get a group homomorphism  $f^* : K^{Cat}(Y, R) \rightarrow K^{Cat}(X, R)$  defined on generators by  $f^*([F]) = [F \circ \Delta f]$ .

The groups  $K^{Cat}(X, R)$  are covariantly functorial in  $R$  by “extension of scalars”. Furthermore,  $X \mapsto K^{Cat}(X, R)$  gives a contravariant homotopy functor, as by definition of concordance it is clear that if  $f : X \rightarrow Y$  is homotopic to  $g$  then  $f^* = g^*$ . In particular, if  $X_0$  is homotopy equivalent to  $X_1$  then  $K^{Cat}(X_0, R) = K^{Cat}(X_1, R)$ .

Let  $S_\bullet^n$  be the singular complex of  $S^n$ .

**Definition 3.6.** For  $k > 0$ , we define

$$K_n^{Cat}(R) = \ker\{(\Delta^0 \rightarrow S_\bullet^n)^* : K^{Cat}(S_\bullet^n, R) \rightarrow K^{Cat}(\Delta^0, R)\},$$

and define

$$K_0^{Cat}(R) = K^{Cat}(\Delta^0, R).$$

In other words,  $K_0^{Cat}(R)$  is the quotient of the free abelian group generated by Morita equivalence classes  $[C]$ , by the subgroup generated by:

$$\{[C_0] - [C_1] + [C_2] \mid \text{there is a short exact sequence } C_0 \rightarrow C_1 \rightarrow C_2\}.$$

To see relationship between the pair of definitions of  $K_n^{Cat}(R)$ , note that for  $X$  a Kan complex a bundle functor  $F : |\Delta(X)| \rightarrow w\mathcal{A}_R$  induces a sequence

$$(3.7) \quad |X| \simeq |\Delta(X)| \xrightarrow{|F|} |w\mathcal{A}_R| \xrightarrow{i} K^{Cat}(R),$$

where  $|X|$  also denotes the geometric realization of  $X$ . And clearly a concordance between  $F_0, F_1$  induces a homotopy between  $i \circ |F_i|$ .

Let  $hMaps(|X|, K^{Cat}(R))$  be the set of free homotopy classes  $[|X| \rightarrow K^{Cat}(R)]$ . As  $K^{Cat}(R)$  is an infinite loop space,  $hMaps(|X|, K^{Cat}(R))$  is naturally an abelian group. Thus we get an induced group homomorphism:

$$(3.8) \quad \tilde{\mathcal{J}}_X : B(X, R) \rightarrow hMaps(|X|, K^{Cat}(R)),$$

induced by the map  $[F] \mapsto [\mathbf{i} \circ |F|]$ .

**Claim 3.9.** *The map  $\tilde{\mathcal{J}}_X$  passes to the quotient, inducing a map*

$$(3.10) \quad \mathcal{J}_X : K^{Cat}(X, R) \rightarrow hMaps(|X|, K^{Cat}(R)).$$

*Furthermore,  $\mathcal{J}_X$  is a group isomorphism, and induces a natural isomorphism  $\mathcal{J}$  from the Ab-valued functor  $X \mapsto K^{Cat}(X, R)$  to the functor  $X \mapsto Maps(|X|, K^{Cat}(R))$ .*

The proof of the claim is postponed, as it involves various algebraic topology details. Note that in general one cannot always concretely present Waldhausen  $K$ -theory groups in this manner. The proof in our setting hinges on the fact that the  $\infty$ -localization of  $\hat{\mathcal{A}}_R$  with respect to Morita equivalences has a concrete model  $\mathcal{M}$ , such that concordance classes of functors  $F : \Delta(X) \rightarrow w\hat{\mathcal{A}}_R$  are in correspondence with homotopy classes  $[X \rightarrow Core(\mathcal{M})]$ , where  $Core(\mathcal{M})$  is the maximal Kan subcomplex. As remarked in the paragraph following Definition 3.1, this  $\mathcal{M}$  is the  $\infty$ -Waldhausen category from which the spectrum  $K^{Cat}(R)$  is constructed.

Given the claim, there is clearly an induced map

$$(3.11) \quad \mathcal{J}_m : K_m^{Cat}(R) \rightarrow \pi_m(K^{Cat}(R)),$$

and it is a group isomorphism.

**3.2. A very basic computation.** We recall the definition of  $K_0(R)$ . This is the quotient of the free abelian group generated by the set of isomorphism classes of projective finitely generated  $R$ -modules, by the subgroup generated by

$$\{[A] - [B] + [C] \mid \text{exists an exact sequence } 0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0.\}$$

If  $\mathcal{E}$  is a triangulated category, one defines  $K_0(\mathcal{E})$  to be the quotient of the free abelian group, generated by the set of isomorphism classes in  $\mathcal{E}$ , by the subgroup generated by

$$\{[A] - [B] + [C] \mid \text{exists a distinguished triangle } A \rightarrow B \rightarrow C \rightarrow A[1]\}.$$

It is known that  $K_k(D^{perf}(Mod_R)) \simeq K_k(R)$ , where  $Mod_R$  denotes the category of projective finitely generated  $R$ -modules and  $D^{perf}(Mod_R)$  denotes the derived category of perfect complexes.

If  $D_n(Mod_R)$  denotes the  $n$ -periodic derived category, we no longer have that  $K_0(R) \simeq K_0(D_n(Mod_R))$ , as there are certain corrections. However we have the following.

**Theorem 3.12** (Saito [18]). *For  $n$  even, the natural homomorphism:*

$$(3.13) \quad \psi : K_0(D_n(Mod_R)) \rightarrow K_0(R)$$

$$(3.14) \quad \psi([V]) = \sum_{i=0}^{i=n} (-1)^i [H^i(V)],$$

*is an isomorphism.*

There is a group homomorphism

$$\phi : K_0^{Cat}(R) \rightarrow K_0(D_2(Mod_R))$$

induced by  $[C] \mapsto [Hoch_\bullet(C)]$  for  $Hoch_\bullet(C)$  the Hochschild homology complex of  $C$ . This readily follows from the fact that the Hochschild functor  $Hoch_\bullet$  is “localizing” which means in this case that a short exact sequence

$$C_0 \rightarrow C_1 \rightarrow C_2$$

in  $\mathcal{A}_R$  is taken to an exact triangle

$$\mathrm{Hoch}_\bullet(C_0) \rightarrow \mathrm{Hoch}_\bullet(C_1) \rightarrow \mathrm{Hoch}_\bullet(C_2) \rightarrow \mathrm{Hoch}_\bullet(C_0)[1],$$

in the derived category  $D_2(\mathrm{Mod}_R)$ , see Keller [10, Theorem 5.2].

**Corollary 3.15.** *The composition map*

$$K_0^{\mathrm{Cat}}(\mathbb{Z}) \xrightarrow{\phi} K_0(D_2(\mathrm{Mod}_{\mathbb{Z}})) \xrightarrow{\psi} K_0(\mathbb{Z}),$$

*is non-trivial.*

*Proof.* Take  $C$  to be a finite type  $\mathbb{Z}_2$ -graded  $A_\infty$  algebra over  $\mathbb{Z}$  such that  $HH_0(C) = \mathbb{Z}$ ,  $HH_1(C) = 0$ ,  $HH_2(C) = \mathbb{Z}$ , where  $HH_\bullet(C)$  is the homology of  $\mathrm{Hoch}_\bullet(C)$ . As a geometric example take  $C = FH_\bullet(L_0, L_0)$  where the latter is the Floer cochain algebra of the equator  $L_0$  in  $S^2$ , then the Hochschild homology  $HH_\bullet(C)$  is isomorphic to the singular homology  $H_\bullet(S^2, \mathbb{Z})$ , [22].

So we get

$$\psi([\mathrm{Hoch}_\bullet(C)]) = 2[\mathbb{Z}] \neq 0 \in K_0(\mathbb{Z}) \simeq \mathbb{Z}.$$

□

A further corollary of the proof above is:

**Corollary 3.16.**  $[\mathrm{Fuk}(S^2, \omega)]$  represents a non-trivial element in  $K_0^{\mathrm{Cat}}(\mathbb{Z})$ .

One natural question is how to produce non-trivial elements in  $K_m^{\mathrm{Cat}}(R)$  for higher  $m$ . We will construct such elements using Hamiltonian fibrations and Floer theory.

#### 4. BUNDLES OF $A_\infty$ CATEGORIES FROM HAMILTONIAN FIBRATIONS

The main construction of [20] has input a smooth Hamiltonian fibration  $M \hookrightarrow P \rightarrow Y$ , with  $(M, \omega)$  a compact, monotone symplectic manifold, e.g.  $\mathbb{CP}^n$ . More generally, a monotone symplectic manifold  $(M, \omega)$  is one satisfying  $[\omega] = \mathrm{const} \cdot c_1(TM)$ , with  $\mathrm{const} > 0$ , as functionals  $\pi_2(M) \rightarrow \mathbb{R}$ .

Denote by  $Y_\bullet$  the simplicial set of smooth maps  $\Delta^n \rightarrow Y$  for  $\Delta^n$  also denoting the standard topological  $n$ -simplex.

The output is a functor

$$(4.1) \quad F_P : \Delta(Y_\bullet) \rightarrow wA_\infty \mathrm{Cat}_R^{\mathbb{Z}_2, \mathrm{strict}},^3$$

that is a bundle functor over  $Y_\bullet$ , whose concordance class  $F_P$  is independent of auxiliary choices (these mainly consist of choices of almost complex structures and Hamiltonian connections.) For  $x : \Delta^0 \rightarrow Y$  a point map  $F_P(x) = \mathrm{Fuk}(P_x, \omega_x)$ ,  $P_x$  denoting the fiber over  $x$ .

We can outline this further: if  $\Sigma : \Delta^1 \rightarrow Y$  is a path from  $x$  to  $y$ , then  $F_P(\Sigma)$  has the set of objects

$$\mathrm{obj} F_P(x) \sqcup \mathrm{obj} F_P(y).$$

For  $L_x \in \mathrm{obj} F_P(x)$  and  $L_y \in \mathrm{obj} F_P(y)$ ,  $\mathrm{hom}_{F_P(\Sigma)}(L_x, L_y)$  is a certain chain complex generated by flat sections of an auxiliary Hamiltonian connection on  $\Sigma^*P$ , with end points of the sections on  $L_x$  and  $L_y$ .<sup>4</sup>

<sup>3</sup>In [20] we work with  $R = \mathbb{Q}$  but this was only used for inverting quasi equivalences. As we don't need to do this here, the construction works over any commutative ring.

<sup>4</sup>These Hamiltonian connections are fixed per simplex, but need to satisfy compatibility conditions involving face maps of simplices.



More generally, for a smooth  $\Sigma : \Delta^n \rightarrow Y$  denote by  $v_i : \Delta^0 \rightarrow Y$ ,  $0 \leq i \leq n$ , the corresponding vertex maps. Then  $F_P(\Sigma)$  has the set of objects

$$(4.2) \quad \sqcup_i \text{obj } F_P(\Sigma)(v_i).$$

The hom sets are defined analogously so that an edge inclusion morphism  $\Sigma^1 \rightarrow \Sigma$  in  $\Delta(Y_\bullet)$  induces a fully-faithful embedding  $F_P(\Sigma_1) \rightarrow F_P(\Sigma)$ . Here by an edge inclusion we mean a morphism of the type:

$$(4.3) \quad \begin{array}{ccc} \Delta^1 & \longrightarrow & \Delta^n \\ & \searrow \Sigma^1 & \downarrow \Sigma_2 \\ & & X. \end{array}$$

The only deep part of the construction is then the definition of the  $A_\infty$  structure on  $F_P(\Sigma)$ . This involves some novel geometric topology of the universal curve, but we will not attempt to outline this further, instead referring the reader to the original construction in [20].

To get the algebraic finiteness condition on the Fukaya category we may impose the following condition.

**Definition 4.4.** We call a compact monotone symplectic manifold of *finite monotone type*, if its monotone Fukaya category (see [22] and also [20]) is finite type.

Let us set  $\widehat{F}_P = T \circ F_P$ . We then have the following corollary of [20, Theorem 6.6].

**Theorem 4.5.** *Let  $M \hookrightarrow P \rightarrow Y$  be a Hamiltonian fibration with fiber  $(M, \omega)$  finite monotone type. Then the assignment*

$$(4.6) \quad [P] \mapsto [\widetilde{\mathcal{J}}_{Y_\bullet}(\widehat{F}_P)] \in h\text{Maps}(Y, K^{Cat}(R)),$$

*is well defined, where  $[P]$  is the Hamiltonian isomorphism class of the fibration  $P$ , and  $\widetilde{\mathcal{J}}$  is as in (3.8).*

As an example, we have:

**Proposition 4.7.**  $\mathbb{CP}^n$  is finite monotone type.

*Proof.* Specifically, we take a concrete model of  $\text{Fuk}(\mathbb{CP}^n)$  with the set of objects given by the orbit of the Clifford torus  $L_0$  by the natural  $\text{Ham}(\mathbb{CP}^n, \omega)$  action. In this case, it is automatic that  $\text{Fuk}(\mathbb{CP}^n)$  is quasi-isomorphic to the  $A_\infty$  algebra  $\text{hom}_{\text{Fuk}(\mathbb{CP}^n)}(L_0, L_0)$ . The Hochschild chain complex of  $\text{hom}_{\text{Fuk}(\mathbb{CP}^n)}(L_0, L_0)$  is quasi-isomorphic to the Hamiltonian Floer chain complex, see for instance [22], which in each degree is finitely and freely generated over  $\mathbb{Z}$ . And so  $\text{Fuk}(\mathbb{CP}^n)$  is finite type for any  $R$ .  $\square$

In particular, taking  $Y = S^m$  we get:

**Corollary 4.8.** *For each commutative ring  $R$ , there are natural homomorphisms:*

$$(4.9) \quad \phi_m : \pi_m(B\text{Ham}(\mathbb{CP}^n)) \rightarrow K_m^{Cat}(R).$$

*And in particular natural homomorphisms:*

$$(4.10) \quad \phi_m : \pi_m(BPU(n)) \rightarrow K_m^{Cat}(R).$$

*Proof.* Let  $\mathbb{CP}^n \hookrightarrow P^U \rightarrow B\text{Ham}(\mathbb{CP}^n)$  be the associated universal bundle.<sup>5</sup> Using Proposition 4.7 together with Theorem 4.5 we get  $\phi_m$  as set maps. To see that they are homomorphisms note that we have a sequence of maps well defined up to homotopy:

$$B\text{Ham}(\mathbb{CP}^n) \xrightarrow{|\widehat{F}_{P^U}|} |w\widehat{\mathcal{A}}_R| \xrightarrow{i} K^{Cat}(R).$$

By the naturality of the assignment  $[P] \mapsto [F_P]$ , see [20, Section 9.2],  $\phi_m$  are the induced maps

$$(i \circ |F_{P^U}|)_* : \pi_m(B\text{Ham}(\mathbb{CP}^n)) \rightarrow \pi_m(K^{Cat}(R)).$$

And so are group homomorphisms. □

**4.1. Generalizing to  $G/T$ .** Given a compact connected Lie group  $G$ , the maximal coadjoint orbit  $G/T$  can be made monotone with respect to the Kirillov-Kostant-Souriau symplectic structure  $\omega_{KKS}$ , which for  $G/T$  is uniquely determined up to deformation equivalence, see for instance [11]. We will from now on omit this monotone  $\omega_{KKS}$  from notation, but implicitly it will always be the chosen symplectic form. If  $\text{Fuk}(G/T)$  was known to be of finite type, we would similarly get homomorphisms  $\pi_m(B\text{Ham}(G/T)) \rightarrow K_m^{Cat}(R)$ . But even in this case without an explicit model for the Fukaya category—like in the case of  $\mathbb{CP}^n$ —these morphisms would be far from concrete. We address both issues as follows.

If  $G = PU(n)$  there is a distinguished monotone Lagrangian torus  $L_{GC}$  of  $PU(n)/T$ , arising as the central fiber of a Gelfand-Cetlin integrable system, see [6, Theorem B]. In a way this extends to a general compact connected  $G$ . In this case,  $G/T$  admits a flat toric degeneration to a toric variety  $X(\Delta)$  associated with a string polytope  $\Delta$ , see [4], [1]. This deformation defines a completely integrable system on  $G/T$ , with moment map to  $\Delta$  [7]. By anchoring the construction to the standard string valuation, one identifies a distinguished toric degeneration where the resulting string polytope is Fano. Within this fixed framework, the fiber over the barycenter of  $\Delta$  is a monotone Lagrangian torus that is well-defined up to Hamiltonian isotopy [5]. This torus will be denoted as  $\mathcal{L}$ . (Not that taking “non-standard” valuations, we may get exotic tori that are not Hamiltonian isotopic to  $\mathcal{L}$  [5].)

Thus we get a finite type full subcategory  $C_{\mathcal{L}} \subset \text{Fuk}(G/T)$ , with the set of objects the orbit of  $\mathcal{L}$  by the  $\text{Ham}(G/T)$  action<sup>6</sup>. We then construct a functor

$$F_{P^U} : \Delta(B\text{Ham}(G/T)_{\bullet}) \rightarrow w\mathcal{A}_R$$

as follows. Denote by  $G/T \hookrightarrow P^U \rightarrow B\text{Ham}(G/T)$  the associated universal bundle. For a point map  $x : \Delta^0 \rightarrow X$  we set  $F_{P^U}(x)$  to be the full subcategory of  $\text{Fuk}(P_x)$ , with the set of objects  $\phi_x(\text{obj } C_{\mathcal{L}})$ , where  $\phi_x : G/T \rightarrow P$  is a frame over  $x$  (a parametrization respecting the structure group). The remaining construction of  $F_{P^U}$  is the same as the general construction of (4.1).

Given this  $F_{P^U}$ , following the proof of Corollary 4.8, we again get maps:

$$(4.11) \quad \phi_{\mathcal{L},G} : \pi_m(B\text{Ham}(G/T)) \rightarrow K_m^{Cat}(R).$$

Composing with the natural map  $BG \rightarrow B\text{Ham}(G/T)$  we also get:

$$(4.12) \quad \phi_{\mathcal{L},G} : \pi_m(BG) \rightarrow K_m^{Cat}(R),$$

and these later maps are in essence concrete, at least within the framework of the representation theory outlined above.

*Remark 4.13.* By the last sentence we mean that the data does not hide anything transcendental unless one counts Hamiltonian perturbations underlying Floer theory (which is used in the construction of  $F_{\mathcal{L}}$ ). But since the perturbations can be made concrete, this is not an issue.

<sup>5</sup>In case the reader is worried about smoothness of this  $P$  (which was an assumption for the bundle functor construction), this isn't an issue as [20] treats it as generalized smooth bundle, for which the construction still goes through.

<sup>6</sup> $C_{\mathcal{L}}$  is possibly not known to be Morita equivalent to  $\text{Fuk}(G/T)$ , so it is just our model in the general sense.

5. HAMILTONIAN ELEMENTS IN CLASSICAL ALGEBRAIC  $K$ -THEORY OF  $R$ 

One of the goals is to obtain Hamiltonian elements in classical algebraic  $K$ -theory of  $R$ . Let  $M \hookrightarrow P \rightarrow Y$  be a Hamiltonian fibration, with compact monotone fiber  $(M, \omega)$ . Let

$$F_P : \Delta(Y_\bullet) \rightarrow wA_\infty \text{Cat}_R^{\mathbb{Z}_2, \text{strict}}$$

be as in the previous section. Then the composition:  $\Phi_P = \text{Hoch}_\bullet \circ F_P$  gives a functor to  $wCh_R^{\mathbb{Z}_2}$ , where  $Ch_R^{\mathbb{Z}_2}$  is the Waldhausen (strict) category of 2-periodic complexes of projective finite rank  $R$ -modules. The Waldhausen  $K$ -theory infinite loop space  $K(Ch_R^{\mathbb{Z}_2})$  will be denoted by  $K^{\mathbb{Z}_2}(R)$ .

Restrict now to  $Y = B\text{Ham}(G/T)$ , and  $P$  the associated universal  $G/T$  bundle over  $B\text{Ham}(G/T)$ . Now by standard homotopy theory,  $|\Delta(Y_\bullet)| \simeq X$ , where for a category  $C$ ,  $|C|$  is shorthand for the geometric realization of the nerve. Then we get a map:

$$|F_P| : B\text{Ham}(G/T) \rightarrow |wCh_R^{\mathbb{Z}_2}|,$$

whose homotopy class is well defined by the fact that the concordance class of  $F_P$  is well defined.

There is a natural map  $|wCh_R^{\mathbb{Z}_2}| \rightarrow K^{\mathbb{Z}_2}(R)$ , by generalities of the Waldhausen construction, see [26, Remark 8.3.2]. Composing with this map we get a natural homotopy class:

$$(5.1) \quad [|F_P| : B\text{Ham}(G/T) \rightarrow K^{\mathbb{Z}_2}(R)].$$

*Proof of Theorem 1.4.* Compose the natural map

$$BG \rightarrow B\text{Ham}(G/T),$$

and the map from (5.1) to obtain  $h$ . □

Of course, at the moment  $h$  is very transcendental, unless we give  $\text{Fuk}(G/T)$  a concrete model as we do in Section 4.1. Or unless we reinterpret the map (5.1) in some more basic terms, as asked in Question 1.

6. LANGLANDS DUAL ALGEBRAIC  $K$ -THEORY ELEMENTS.

It is natural to ask how Langlands duality fits into our story. Denote by  $G^\vee$  the maximal compact subgroup of the Langlands dual  $G_\mathbb{C}^\vee$  of the complexification  $G_\mathbb{C}$ . Via Theorem 1.4, we have natural maps:

$$\begin{aligned} h_G &: \pi_m(BG) \rightarrow K_m^{\mathbb{Z}_2}(R), \\ h_{G^\vee} &: \pi_m(BG^\vee) \rightarrow K_m^{\mathbb{Z}_2}(R). \end{aligned}$$

If  $G$  is simply laced (e.g.  $G = SU(n)$ ) it is not hard to see that the images of these maps are the same. In this case  $G/T = G^\vee/T$  and then the images of  $G$  and  $G^\vee$  in  $\text{Ham}(G/T)$  coincide, so that the assertion readily follows. Motivated by this:

**Proposal 1.** *For each compact Lie group  $G$  the images of  $h_{G,m}$  and  $h_{G^\vee,m}$  coincide.*

Furthermore, there are “lifts”<sup>7</sup> as in (4.12):

$$\begin{aligned} \phi_{\mathcal{L},G} &: \pi_m(BG) \rightarrow K_m^{\text{Cat}}(R), \\ \phi_{\mathcal{L},G^\vee} &: \pi_m(BG^\vee) \rightarrow K_m^{\text{Cat}}(R), \end{aligned}$$

and we can again ask if the images coincide. For  $m = 0$ , this is saying that  $\text{Fuk}(G/T)$  and  $\text{Fuk}(G^\vee/T)$ <sup>8</sup> represent the same classes in  $K_0^{\text{Cat}}(R)$ . This assertion holds if these categories were Morita equivalent,

<sup>7</sup>Given Keller’s theorem outlined in Section 3.2, they may be lifts in the formal sense, that is the Hochschild functor should induce a map of spectra  $\mathfrak{h}$ , such that  $\phi_{\mathcal{L},G} \circ \mathfrak{h} = h_G$ . However we do not verify this.

<sup>8</sup>Or to be more restrictive their models  $C_{\mathcal{L}}$  as in Section 4.1.

but it is much weaker. The case  $m = 0$  is then loosely comparable to the homological mirror symmetry conjecture of Kontsevich.

Note that the domains of  $\phi_{\mathcal{L},G}$  and  $\phi_{\mathcal{L},G^\vee}$  and the manifolds  $G/T$  and  $G^\vee/T$  are generally very different. The idea is that the images in (categorified) algebraic  $K$ -theory depend only on some representation theoretic data that is invariant under Langlands duality. Thematically, such a phenomenon already appears in electric-magnetic dualities, and this is one inspiration. A related work on topological aspects of Langlands duality/correspondence is [3]. The seminal reference for electric-magnetic duality vis-à-vis Langlands duality is Kapustin-Witten [8].

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